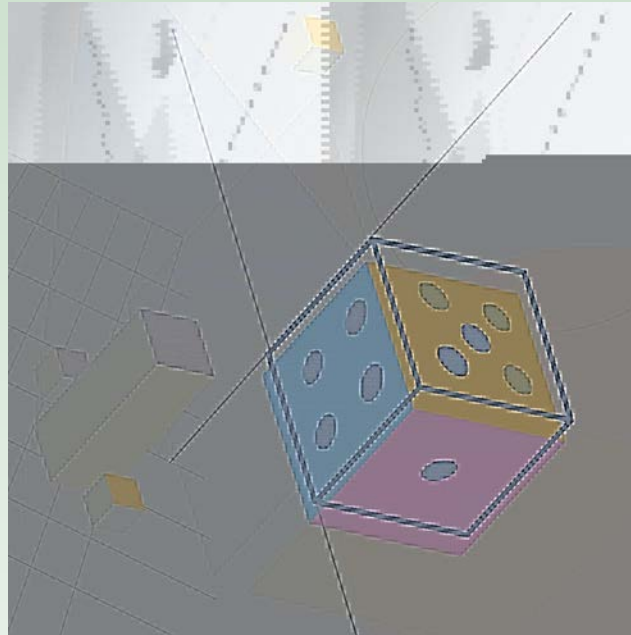


PROBABILITY THEORY

**2019-2020 SPRING SEMESTER
MIDTERM**

GR:2

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1. The probability density function of X is given below. If $E(X) = 3/5$, find the values of a and b .

$$f(x) = \begin{cases} a + bx^2, & 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

▲ $a=3/5, b=6/5$

◆ $a=6/5, b=3/5$

● $a=-2, b=3$

■ There is no enough information to find the values.

2. A box contains 4 white and 3 black balls, while a second box contains 3 white and 5 black balls. A ball is randomly selected from the first box and is thrown into the second box without seeing its color. Accordingly, what is the probability that the ball selected from the second box is black?

▲ $25/63$

● $7/18$

◆ $11/18$

■ $38/63$

3. For the events A and B, $P(A|B) = 1/4$, $P(B|A) = 1/2$ and $P(A \cap B) = 3/16$. According to this, which of the below(s) are correct?

▲ A and B are independent and disjoint events.

◆ A and B are independent, but not disjoint events.

● A and B are dependent, but not disjoint events.

■ A and B are dependent and disjoint events.

4. Which of the below(s) are correct?

I) If $A \subset B$, then $P(B|A) = 1$.

II) If $A \cap B = \emptyset$, then $P(A|B) = 0$.

III) If X_1 and X_2 are independent $E(X_i) = \mu_i, Var(X_i) = \sigma_i^2$ and $Y = X_2 - X_1$, then $E(Y) = \mu_2 - \mu_1, Var(Y) = \sigma_2^2 + \sigma_1^2$.

IV) If $cov(X_1, X_2) = 0$, then X_1 and X_2 are definitely independent.

V) $Var(cX + b) = cVar(X) + b$

▲ I, II, III and IV

● II and III

◆ I, III and IV

■ I, II and III

5. How many different numbers with 7 digits can be written by using 2,2,2,1,1,0,0 ?

▲ 100

● 210

◆ 150

■ 84

6. If the moment generating function of a random variable X is given below, find its expected value and variance $(-1/2 < \lambda < 0)$.

$$M_X(t) = 1 - e^{\lambda(e^t - 1)}, \quad (-1/2 < \lambda < 0)$$

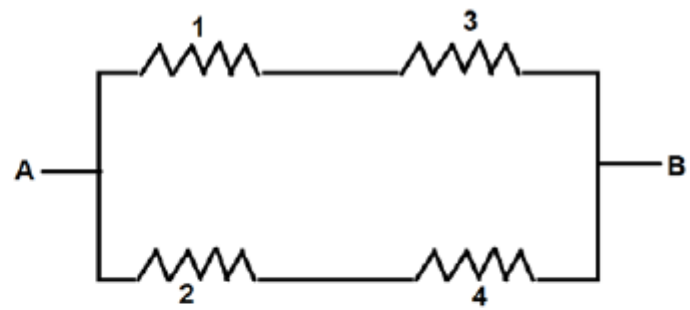
▲ $E(X) = \lambda, \text{Var}(X) = \lambda$

● $E(X) = -\lambda, \text{Var}(X) = -\lambda - 2\lambda^2$

◆ $E(X) = -\lambda, \text{Var}(X) = -\lambda$

■ $E(X) = -\lambda, \text{Var}(X) = -\lambda + 2\lambda^2$

7. In the adjacent electric circuit, the current flows if the electric switch is closed. The switches operates



independently. The probability that switch (1) and (3) are open is separately 0.3, and switch (2) and (4) are open is separately 0.4. Find the probability that the current flows from A to B.

▲ $(0.3)^2 + (0.4)^2 - (0.3)^2(0.4)^2$

◆ $(0.7)^2 + (0.6)^2 - (0.7)^2(0.6)^2$

● $(0.7)^2 + (0.6)^2$

■ $(0.3)^2 + (0.4)^2$

8. If $E(X) = 0$ and $Var(X) = 1$, then find the lower bound of the probability $P(-2 < X < 2)$.

▲ $3/4$

● $1/2$

◆ $1/6$

■ $1/4$

Questions #9-11 are based on the following information:

The cumulative distribution of a random variable X is given below.

$$F(x) = \begin{cases} 0, & x < 1 \\ \ln x, & 1 \leq x \leq e \\ 1, & x > e \end{cases}$$

9. What is the probability density function $f(x)$ and the probability $P(2 < X \leq 2.3)$?

▲ $f(x)=1/x^2, \ln(2,3)-\ln 2$

◆ $f(x)=1/\ln x, \ln(2,3)-\ln 2$

● $f(x)=1/x, 1-\ln 2$

■ $f(x)=1/x, \ln(2,3)-\ln 2$

The cumulative distribution of a random variable X is given below.

$$F(x) = \begin{cases} 0, & x < 1 \\ \ln x, & 1 \leq x \leq e \\ 1, & x > e \end{cases}$$

10. What is the expected value of X ?

▲ $E(X)=1$

● $E(X)=e-1$

◆ $E(X)=1-e$

■ $E(X)=e^2-1$

The cumulative distribution of a random variable X is given below.

$$F(x) = \begin{cases} 0, & x < 1 \\ \ln x, & 1 \leq x \leq e \\ 1, & x > e \end{cases}$$

11. What is the variance of X ?

▲ $\text{Var}(X) = 2e - (e^2 + 3)/2$

● $\text{Var}(X) = (e^2 - 1)/2$

◆ $\text{Var}(X) = 1 - e$

■ $\text{Var}(X) = (e^2 + 3)/2$

Questions #12-14 are based on the following information:

The joint probability function of the continuous random variables X and Y is given below.

$$f(x, y) = \begin{cases} 4xy, & 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

12. Find the values of $E(X)$ and $Var(X)$, respectively.

▲ $E(X)=2/3, Var(X)=1/2$

● $E(X)=1/18, Var(X)=1/2$

◆ $E(X)=2/3, Var(X)=1/18$

■ $E(X)=1/2, Var(X)=2/3$

The joint probability function of the continuous random variables X and Y is given below.

$$f(x, y) = \begin{cases} 4xy, & 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

13. Find the value of $\text{cov}(X, Y)$.

▲ $4/9$

◆ $2/3$

● $1/2$

■ 0

The joint probability function of the continuous random variables X and Y is given below.

$$f(x, y) = \begin{cases} 4xy, & 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

14. Find the values of $E(X + Y)$.

▲ $4/9$

● $4/3$

◆ $-4/3$

■ 0

Questions #15-20 are based on the following information:

The joint probability function of the discrete random variables X and Y is given below.

$$f(x, y) = \begin{cases} k(x + 2y), & x = 2, 3; \ y = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$$

15. Find the values of k .

▲ 1/14

● 1/12

◆ 1/13

■ 1/11

The joint probability function of the discrete random variables X and Y is given below.

$$f(x, y) = \begin{cases} k(x + 2y), & x = 2, 3; y = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$$

16. Find the values of $P(X > 1, Y \leq 1)$ and $F(2, 1)$, respectively.

▲ 1, 5/7

◆ 1, 3/7

● 3/7, 3/7

■ 3/7, 6/7

The joint probability function of the discrete random variables X and Y is given below.

$$f(x, y) = \begin{cases} k(x + 2y), & x = 2, 3; y = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$$

17. If $g(x)$ and $h(y)$ are respectively the marginal distributions of the random variables X and Y , determine the values of $g(3)$ and $h(0)$, respectively.

▲ 4/14, 5/14

● 3/7, 9/14

◆ 9/14, 5/14

■ 4/7, 5/14

The joint probability function of the discrete random variables X and Y is given below.

$$f(x, y) = \begin{cases} k(x + 2y), & x = 2, 3; y = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$$

18. Find the values of conditional distribution of $X = 2$ and $X = 3$, respectively, If it is known that $Y = 1$.

▲ $4/9, 5/9$

● $4/7, 3/7$

◆ $1/3, 2/3$

■ $3/5, 2/5$

The joint probability function of the discrete random variables X and Y is given below.

$$f(x, y) = \begin{cases} k(x + 2y), & x = 2, 3; \ y = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$$

19. Compute the conditional mean (expected value) of X given $Y = 1$.

▲ $2/9$

● $15/9$

◆ $23/9$

■ $8/9$

The joint probability function of the discrete random variables X and Y is given below.

$$f(x, y) = \begin{cases} k(x + 2y), & x = 2, 3; \ y = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$$

20. Compute the conditional variance of X given $Y = 1$.

▲ 526/81

◆ 61/81

● 20/81

■ 400/81